Oracle Database (on-premise) Migration To AWS RDS Oracle

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Jyoti Mann

This white paper describes strategies on how to migrate an on-premises Oracle database to Amazon Relational Database Service (Amazon RDS) for Oracle, using AWS Database Migration Service (AWS DMS) to migrate the data and AWS Schema Conversion Tool (AWS SCT) to convert the database schema.

The instruction in the whitepaper written to help organizations as a simple migration of on-premise single instance of Oracle database to over to RDS Oracle. This white paper will cover:

- Features and Benefits of AWS RDS
- Steps and strategy of Migration including:
- Migration Planning and work load qualification
- Schema Migration
- Data Migration

This document has specifically referred to AWS recommended guide and steps of migration.

Features and Benefits of AWS RDS

Amazon RDS is a fully managed service for running relational databases on Amazon Web Services (AWS). It has support for six different database engines, including commercial options such as Microsoft SQL Server and Oracle.

Amazon RDS is a cloud native service where most of the database operations are managed by AWS. Having day to day database management task taken care by AWS, Database administration and IT support team can focus on innovation and more valuable task. Amazon RDS handles instance failover, data backups, and software updates, so you can enjoy

- The efficiencies of the AWS Cloud for a fast and reliable database option.
- Pain free scalability
- Higher availability
- Cost optimisation on both software licences as well as infrastructure costs.

NOTE: Amazon RDS Oracle at present doesn't offer RAC capability, however you can use FLASHGRID to enable RAC capability.

Workload Qualifying

Having a business case ready and approval of stakeholders, we now move to the next step of qualifying the workload of migration. This can be done in many ways, however this whitepaper recommends to work with AWS WQF (workload qualifying framework). WQF includes self-service tools, a questionnaire, and design review guidance. AWS solutions architects, partners, and consultants can use these to classify workloads to properly determine the magnitude of migration, staff-hour consumption, and to which AWS service to migrate. The WQF classifies two main types of database workloads—OLTP and DW—into five separate categories, to determine the effort that is required when moving the data.

Category	Workload Type	Ease of migration	Man-hour consumption
1	ODBC/JDBC workload	Easy	Light
2	Light Oracle feature workload	Easy	Medium
3	Heavy Oracle feature workload	Hard	Heavy
4	Oracle-specific application framework	Difficult to migrate	N/A
5	OCI workloads	Hard	Heavy

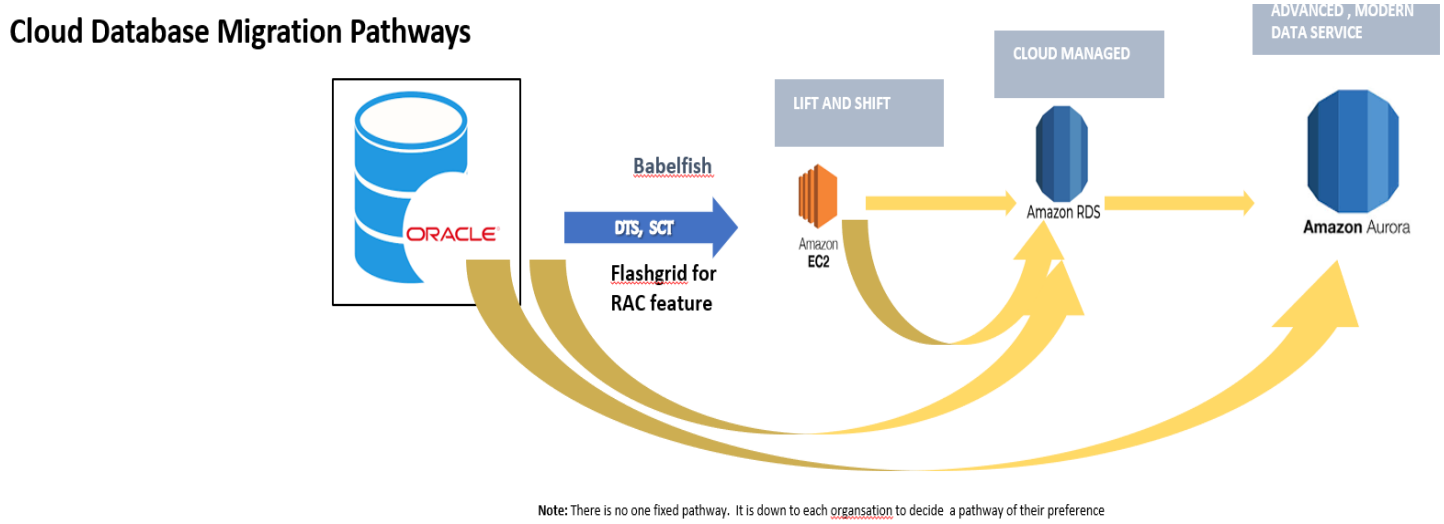
Migration pathways and options:

At this stage it is equally important to understand the full business case. For example, building a business case and listing business current pain points. Once you have a full detail of your Oracle on premise environment as in database numbers, version, data size, data attributes and what is target database RTO and RPO etc. We can work on the migration using Oracle native tools to third party tools e.g. data pump, RMAN backups.
Depending on our data size we will determine our data migration strategy as in do we need a snowmobile or using direct connect over a VPN will be
sufficient. We have many questions to consider

Do we require zero downtime migration?
Do we require data to be encrypted in transition and at rest?
What is the total size of our database?
Required RTO and RPO?
Do we require a Read instance?
Do we require a DR instance?

What Network options we have available and would it sufficient to transfer the data in given migration window?

In terms of pathways many there are few options:



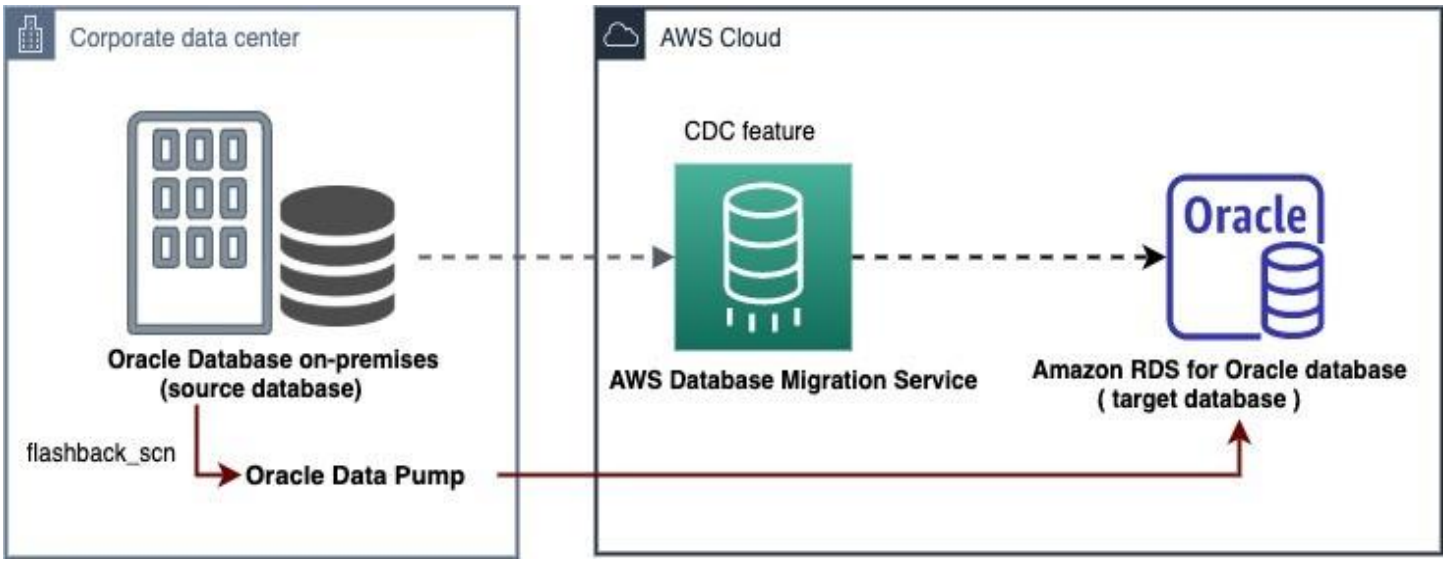


Fig1: On-premise Oracle database migration to AWS RDS Oracle with point in time using DMS CDC.

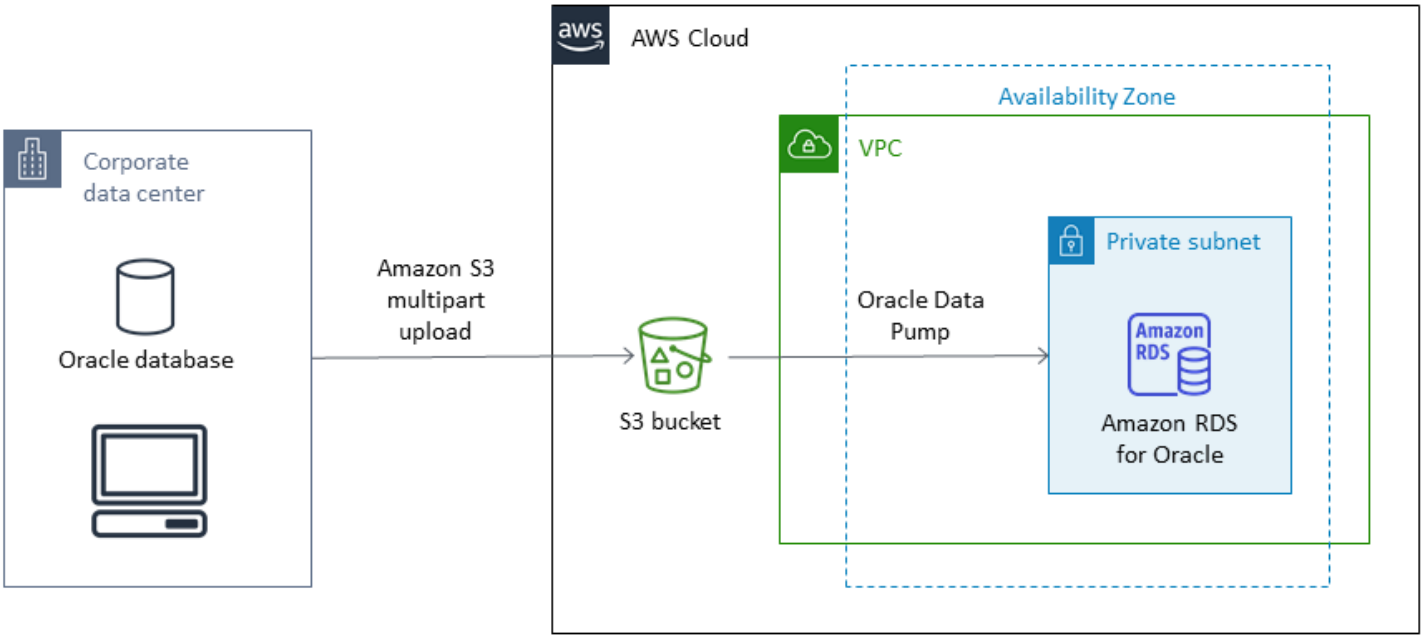


Fig2: On-premise Oracle database migration to AWS RDS Oracle using Oracle native tool Datapump

Planning the migration

The planning your migration work is the key stage and one you should not skip it. It will assist you in the entire migration process to make it more efficient and smooth the entire migration process. It will help you identify which databases to migrate and how, to define the environments required to develop the correct migration approach.

1. **Planning Stage 1:** Capturing the criteria of the migration. This stage you must capture what are the essential criteria for the migration. For example, what is the business downtime options, is it a real-time or there is a downtime window available? If yes, what is the duration of downtime window? Does your data to be encrypted during the migration? What is security and compliance requirement of your organisation during the migration.
2. **Planning stage 2 :** Migration process and infrastructure considerations. This stage you will identify source server infrastructure setup. It also includes identifying and selecting the migration process, to list required access on the source to target database and required infrastructure hardware and pre-requisite of tools and software for the migration. Always work with pre-created template to make a recording of all your findings. This whitepaper strongly advises to document all your work. It really helps to go back and refer to your work and identifying root cause
3. Analysis in case of a future issue or challenge.
4. Validate the versions of the source and target databases.
5. Identify the hardware requirements for the target server instance.

6. Identify storage requirements (storage type and capacity).
7. Choose the proper instance type based on capacity, storage features, and network features.
8. Identify the network access security requirements for the source and target databases.
9. Identify the application migration strategy.

3. **Planning Stage 3**: Building your test (source and target) environment, creating a playbook of your migration before applying it to your production. Testing your migration is not only essential but it assists you ironing al the challenges and issues in the test environment and allows a smooth and zero impact final production implementation.
4. Create a virtual private cloud (VPC).
5. Create security groups.
6. Configure and start an Amazon RDS DB instance.
7. Creating DMS instance and task

The whitepaper discusses the migration process steps using AWS DMS, SCT and WQF.

- Setup tools for each different stage of the migration process
- Appropriate instance types and database versions for the source environment, migration tools, and target environment.
- Building and testing the environment for Oracle, AWS DMS, and RDS Oracle.

- Before starting the actual migration. This white paper recommends you to take the full inventory of your on-premise oracle database and deprecate any objects that are no longer needed. Also, purge any historical data that you no longer need.

- move flat files and long strings stored in LOBs, CLOBs, LONGs, and so on into Amazon S3 or Amazon Dynamo DB. This process requires client software changes, but it reduces the complexity and size of the database and makes the overall migration process more efficient. DMS has its limitation and is not able to migrate LOBS values.

- The objects, like PL/SQL packages and procedures, need to be manually migrated using SCT.

Having done all the above. Let's start with the migration:

Migration are divided into five steps:

- Create an Oracle database instance in Amazon RDS
- Create a replication instance in AWS Database Migration Service (DMS)
- Create endpoints in AWS DMS.
- Create a replication task in DMS.
- Complete the migration and cleanup process.

Note : listed steps for AWS Schema conversion tool (SCT) for a case where you are performing a heterogeneous migration and data conversion may be required , for example Oracle to RDS Postgres SQL.

Heterogeneous data schema conversion: Create your schema in the target database.

During this stage we analyze the schema that you want to migrate. We use the AWS Schema Conversion Tool (AWS SCT) to perform the analysis. We start with creating a new project, set source as the Oracle database and the target being AWS RDS PostgreSQL. When you're connected, select the name of the schema that you want to migrate on the left side. Right-click the schema name and choose Convert Schema. Create and then View / Assessment Report View. The AWS SCT assessment report is a high-level overview of the effort required to convert your Oracle database to PostgreSQL. For further information refer to Documentation for AWS SCT and AWS DMS, DMS Step-by-Step Walkthroughs, DMS Best Practices

Step 1: Create an Oracle database instance in Amazon RDS

To get started, navigate to the Amazon RDS console. On the main page, choose Create database to create a new database.

In the Engine options section, choose Oracle as your Engine type. Then choose the edition and version of Oracle you want to use.

When using Oracle in Amazon RDS, you can choose whether to bring your own license or to create a database instance with the commercial license included.

The database creation wizard includes templates to make it easier to configure the settings for your Amazon RDS database. If you are creating this database to be used in production, you should choose the production template.
In the Settings section, give your database a name, and set the master user name and password. Make sure you write these down because you need them to connect to your database and create additional users.

Next, choose the database instance size. You should make this choice based on your estimated capacity.

If you want to increase or decrease your database instance size in the future, Amazon RDS allows you to do that easily. However, you may incur some downtime to change your database size.

Next, configure the storage options for your Amazon RDS database. There are two storage options in Amazon RDS: general purpose and provisioned IOPS, or I/O operations per second. With general-purpose storage, you receive 3 IOPS per GiB of storage allocated. Thus, 100 GiB of storage would have 300 IOPS. Additionally, you receive burst capacity up to 3,000 IOPS.

With provisioned storage, you provision IOPS separately from your storage capacity. This allows you to tune your storage and operations settings to fit your needs.

There are some additional configuration options you can configure, including settings on backups, monitoring, maintenance, and automated upgrades. The default settings work for most situations, but you should review them to ensure they work for your needs.

The end of the database creation wizard shows you the estimated monthly costs for your database instance.
Choose Create database to create your database instance.

As Amazon RDS is provisioning your infrastructure and initializing your database, the Status of your database is Creating.

When your database is ready to use, its Status is Available.

Step 2: Create a replication instance in AWS Database Migration Service (DMS)

To create a replication instance, go to the Replication Instances section of the AWS DMS console. Choose Create replication instance to begin the replication instance creation wizard.

In the Replication instance configuration section, give your replication instance a name and description. Then choose your instance class. The instance class you use depends on the size of your existing database and the amount of data flowing through it.

Then choose an engine version for AWS DMS. Finally, choose the amount of allocated storage for your replication instance.

As you continue in the Replication instance configuration section, you need to choose a VPC for your replication instance. Choose the same VPC in which you provisioned your Amazon RDS database to ease network access for the replication instance.

You may choose to have a Multi-AZ setup for your replication instance for redundancy. If you will use AWS DMS to keep two databases in sync over a long period of time, you may want to use a Multi-AZ setup. If you are performing a one-time migration of your data from an existing database to a fully managed database in Amazon RDS, you likely don't need a Multi-AZ setup.

Finally, choose whether your replication instance should be publicly accessible. If your existing database is in the same VPC as your new database and your replication instance, you don't need your replication instance to be publicly accessible. If not, you need your replication instance to be publicly accessible.

Next, open the Advanced security and network configuration section. For the VPC security group(s) configuration, choose the same security group that you attached to your Amazon RDS database. This allows your replication instance to access your Amazon RDS database.

You may also edit the maintenance and tags settings.

When you're ready, choose Create to create your replication instance in AWS DMS.

After you choose Create, AWS provisions your replication instance. It shows a Status of the replication instance is Creating while AWS provisions and initializes your instance.

When your replication instance is ready to go, its Status is Available.

While you are waiting for your replication instance to be available, go to the Security Groups section in the Amazon EC2 console. You need to add a rule to your security group to allow your replication instance to access your database. In the Security Groups section, find the security group you attached to your Oracle database instance and your replication instance, and choose it.

Your security group has an existing rule that allows for access to your Oracle instance from the IP address you used to create the database. Remove the existing IP address and enter the name of the security group used for your Amazon RDS database instance and replication instance.
Your screen should look as follows.
Choose Save rules to save the updated rules for your security group.
When your replication instance is available and you have updated the rules for your security group, you may move on to the next module.

Step3: Create endpoints in AWS DMS.

A replication task is a job to migrate data from one database to another by using AWS DMS. Before creating a replication task, you must register endpoints for your source and target databases. An endpoint describes the connection address, credentials, and other information required to connect to a database.

Here we create the endpoint for your target database. This is the database you created in Amazon RDS.

Navigate to the Endpoints section of the AWS DMS console. Choose Create endpoint to create a new endpoint.

In the endpoint creation wizard, choose to create a Target endpoint. Choose Select RDS DB instance, and choose your newly created Amazon RDS database in the dropdown
Before you save your endpoint, test the connection to ensure that it was configured correctly. To test it, open the Test endpoint connection section.
Choose the replication instance you want to use, and then select Run test. After a few seconds, you should see a Status of successful. This indicates that you configured your security group and endpoint correctly. To save your endpoint, select Create endpoint.

Follow these steps again to create an endpoint for your source database. Because the source database is not hosted in Amazon RDS, you need to fill out the connection endpoint, port, and credentials yourself.

You also need to ensure that your replication instance has network access to your source database. If your source database is hosted on Amazon EC2, allow traffic from your replication instance security group into the source database security group. If your source database is not hosted on Amazon EC2, you need to handle the network settings according to the location of your source database.

Before moving on to the next module, you should have two endpoints configured: one for your source database and one for your target database. Make sure that you have tested both endpoints and can successfully connect to both databases. Then proceed to the next module.

Step4: Create a replication task in DMS.

A replication task is responsible for migrating data from a source database to a target database. In your case, you are moving data from an existing database to your newly created database in Amazon RDS.

To get started, navigate to the Database migration tasks section of the AWS DMS console. Choose Create task to create a new replication task.

In the Task configuration section, set up the parameters of your replication task. Give your task a name, and select the replication instance you created in an earlier module. Then choose the source endpoint for your existing database and your target endpoint for your fully managed database in Amazon RDS.

You need to choose a migration type. There are two migration types:

1. Migrate existing data, which performs a one-time process to copy data from your source database to your target database.
2. Replicate ongoing changes, which copies all ongoing operations from your source database to your target database.

If you are migrating your application from using a self-managed database to using a fully managed database, you want to use both types. The first type copies all data in your database, and the second type ensures that all additional updates are replicated to your new database until you switch your application to use the new database.

For the migration type, choose Migrate existing data. Note that this requires you to have logical replication enabled on your source database.

In the Table mappings section, select the tables you want to copy over. Enter the name of the schemas and tables you want to copy. You can use % as a wildcard character to copy multiple tables or schemas.

When you are ready, choose Create task to start your replication task.

After you create your task, it is shown in the Database migration tasks section with a Status of Creating.

After the migration of existing data is complete, it shows a status of Load complete, replication ongoing. Any updates to your source database at this point are copied to your target database.

After performing a migration, you should ensure that the migration completed successfully. This includes performing row counts in both the source and target database as well as viewing records in both databases to ensure they conform. If you are performing ongoing replication, you should ensure that changes to your source database are replicated to your target database.

Closing the migration process

Below I have listed some task you must perform to close the project.

- Shut down the temporary AWS resources.
- Review and validate the project documents.
- Gather metrics for time to migrate, percentage of manual vs. tool, cost savings, etc.
- Close out the project and provide feedback.

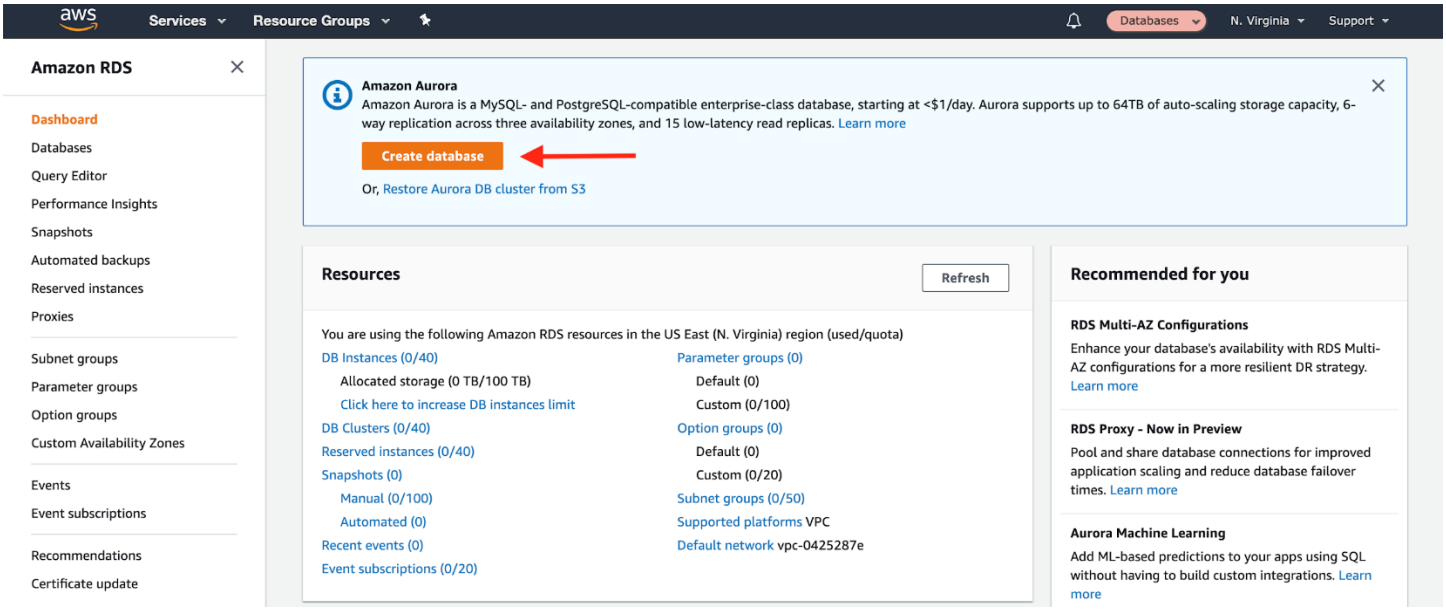
Summary

Migrating Oracle database to AWS RDS Oracle has many benefits. Migration can be a complex process depending on your business constraints and data complexity. However, a well-planned migration assists in making the whole process smooth, risk free and avoid unexpected issues and failures. Working with right migration tools as in AWS DMS and SCT makes the migration simpler and easy to deal with complex data conversion to capturing and applying data changes. Thorough testing and practice the migration steps outlined in this paper, performing POC and performing thorough testing is the recommended approach and it will all contribute in a successful final production implementation.

Final Note: Please refer to the Appendix for images of the migration steps.

Appendix

- Creating a target database AWS RDS Oracle instance



Engine options

Engine type Info

- Amazon Aurora
- MySQL
- MariaDB
- PostgreSQL
- Oracle
- Microsoft SQL Server

Edition

- Oracle Enterprise Edition
  Efficient, reliable, and secure database management system that delivers comprehensive high-end capabilities for mission-critical applications and demanding database workloads.
- Oracle Standard Edition
  Affordable and full-featured database management system supporting up to 32 vCPUs.
- Oracle Standard Edition One
  Affordable and full-featured database management system supporting up to 16 vCPUs.
- Oracle Standard Edition Two
  Affordable and full-featured database management system supporting up to 16 vCPUs. Oracle Database Standard Edition Two is a replacement for Standard Edition and Standard Edition One.

Version Info

Oracle 19.0.0.0.ru-2020-01.rur-2020-01.r1

License

license-included

Settings

DB instance identifier Info

Type a name for your DB instance. The name must be unique cross all DB instances owned by your AWS account in the current AWS Region.

my-database

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens (1 to 15 for SQL Server). First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username Info

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. First character must be a letter

☐ Auto generate a password

Amazon RDS can generate a password for you, or you can specify your own password

Master password Info

••••••••

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), "(double quote) and @ (at sign).

Confirm password Info

••••••••

DB instance size

DB instance class Info

Choose a DB instance class that meets your processing power and memory requirements. The DB instance class options below are limited to those supported by the engine you selected above.

◉ Standard classes (includes m classes)

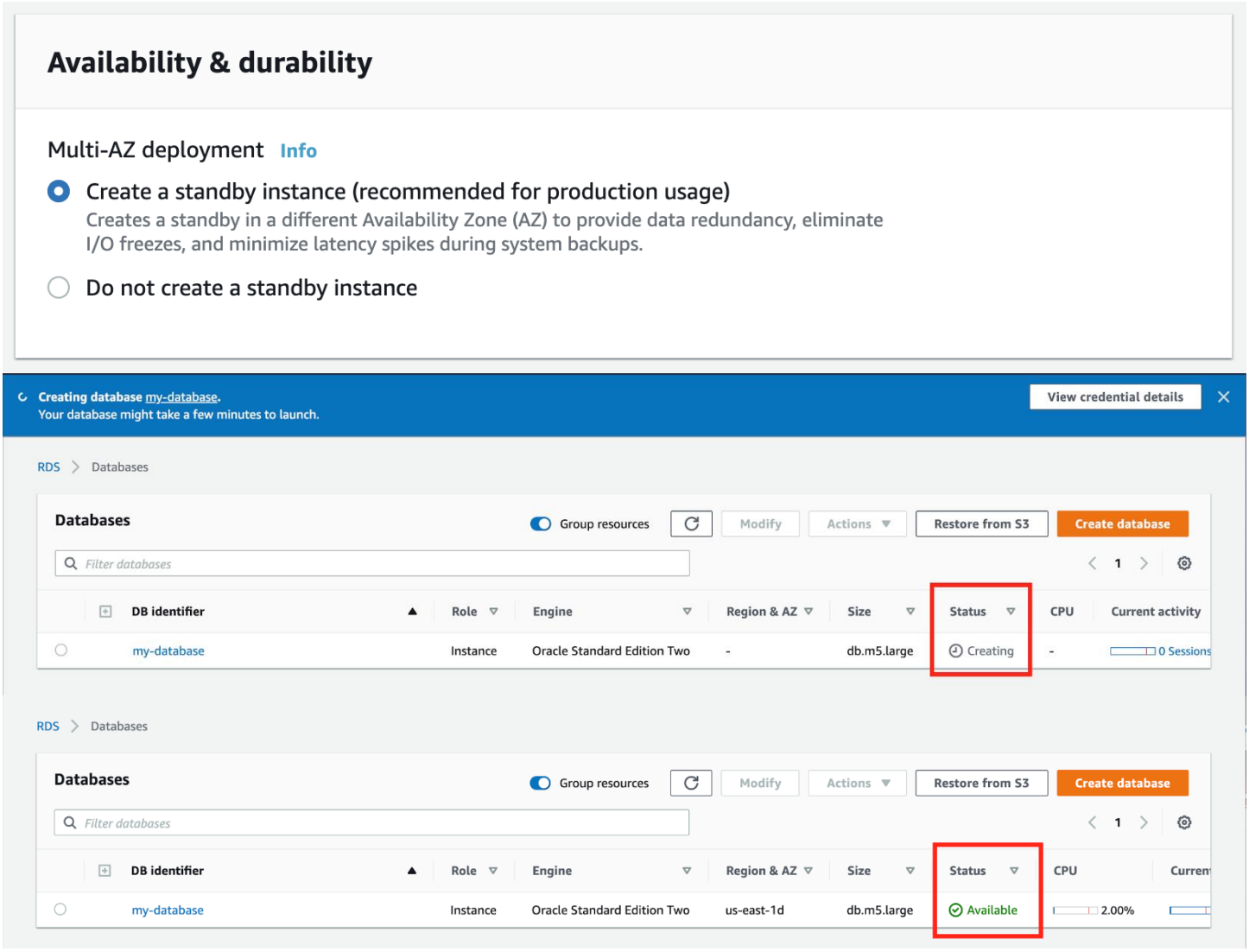
○ Memory Optimized classes (includes r and x classes)

○ Burstable classes (includes t classes)

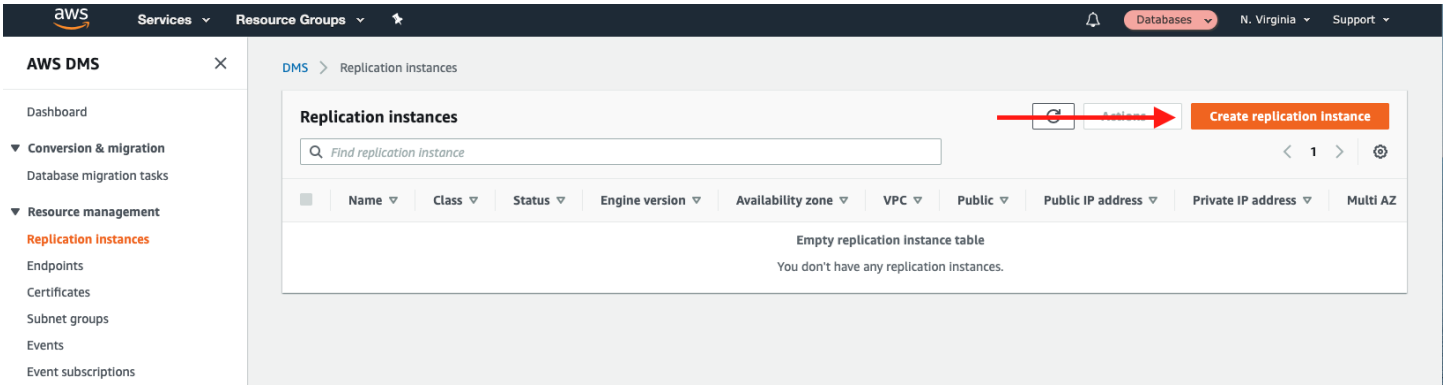
db.m5.large ▼
2 vCPUs 8 GiB RAM EBS: 3500 Mbps

◯ Include previous generation classes

▶ Additional configuration - Optional



➢ Creating Replication task in DMS



Replication instance configuration

Name
The name must be unique among all of your replication instances in the current AWS region.

my-replication-instance

Replication instance name must not start with a numeric value

Description

Replication instance for moving to managed Oracle on RDS

The description must only have unicode letters, digits, whitespace, or one of these symbols: _.:/=+-@. 1000 maximum character.

Instance class
Choose an appropriate instance class for your replication needs. Each instance class provides differing levels of compute, network and memory capacity.

dms.t2.medium

Billing is based on DMS pricing.

Engine version
Choose an AWS DMS version to run on your replication instance.

3.3.1

Allocated storage (GiB)
Choose the amount of storage space you want for your replication instance. AWS DMS uses this storage for log files and cached transactions while replication tasks are in progress.

50

VPC
Choose an Amazon Virtual Private Cloud (VPC) where your replication instance should run.

vpc-0425287e

☐ Multi AZ
If you choose this option, AWS DMS will perform a multi-AZ deployment, with a primary instance in one availability zone (AZ) and a standby instance in another AZ. This configuration provides a highly available, fault-tolerant replication environment.

Billing is based on DMS pricing.

☐ Publicly accessible
If you choose this option, AWS DMS will assign a public IP address to your replication instance, and you'll be able to connect to databases outside of your Amazon VPC.

Advanced security and network configuration

Replication subnet group

Choose a subnet group for your replication instance. The subnet group defines the IP ranges and subnets that your replication instance can use within the Amazon VPC you've chosen.

default-vpc-0425287e

Availability zone

Choose an availability zone (AZ) where you want your replication instance to run. The default is "No preference", meaning that AWS DMS will determine which AZ to use.

No Preference

VPC security group(s)

Choose one or more security groups for your replication instances. The security group(s) specify inbound and outbound rules to control network access to your replication instance.

Use default

oracle-database

KMS master key Info

(Default) aws/dms

Account

Description

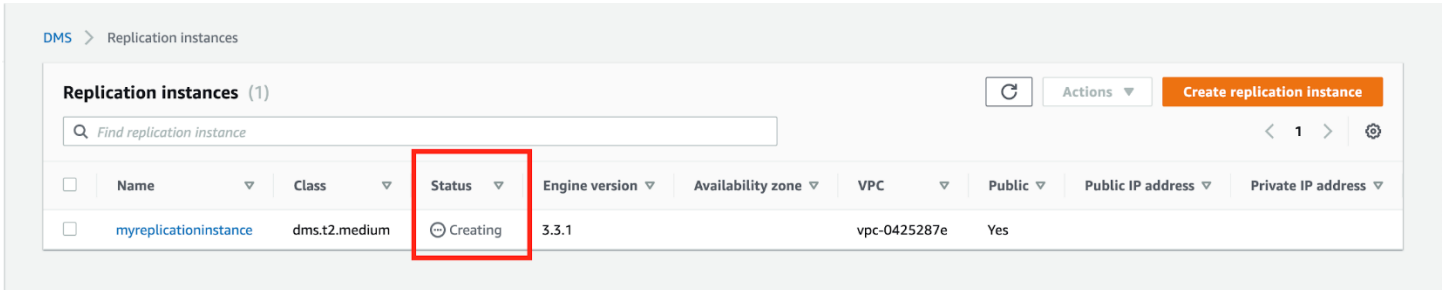
Key ARN

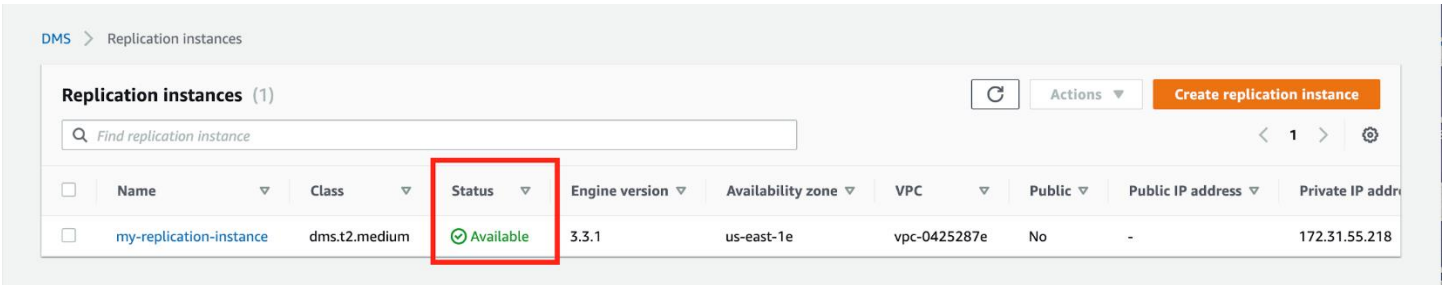
Advanced security and network configuration

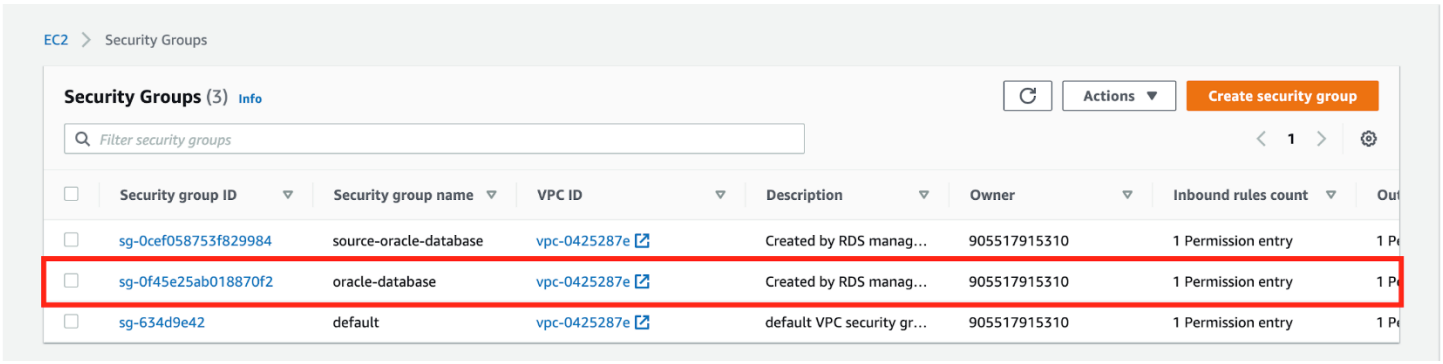
Maintenance

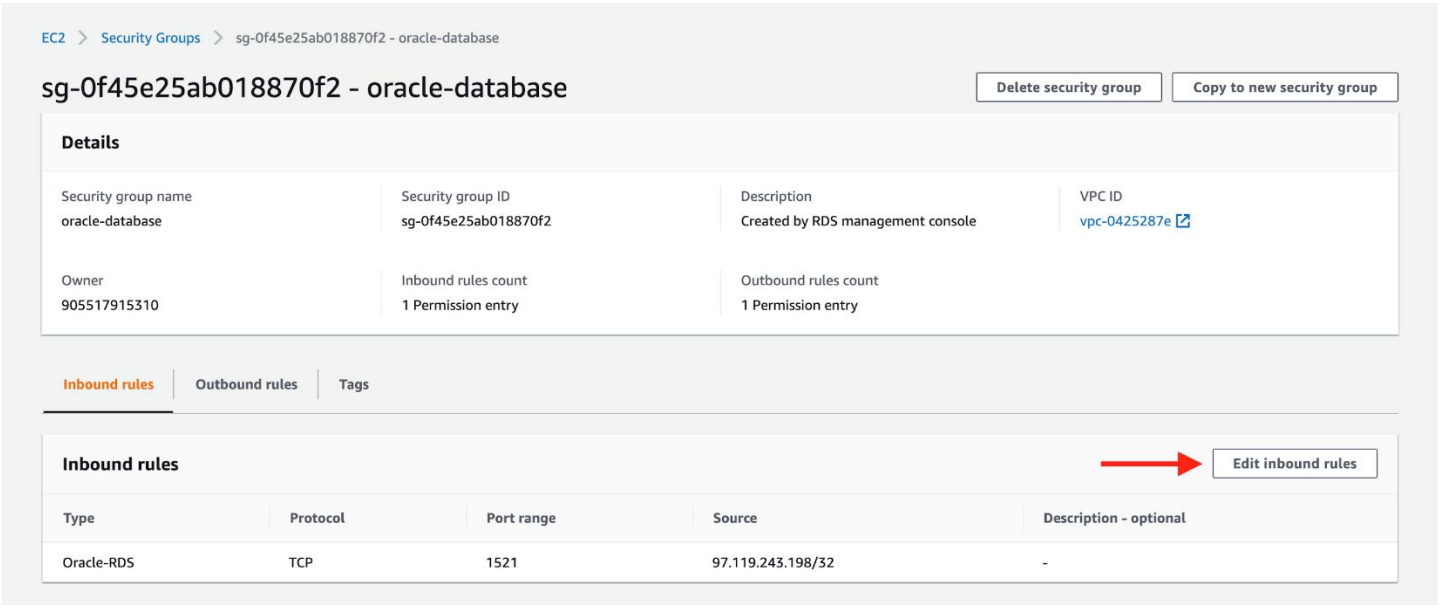
Tags

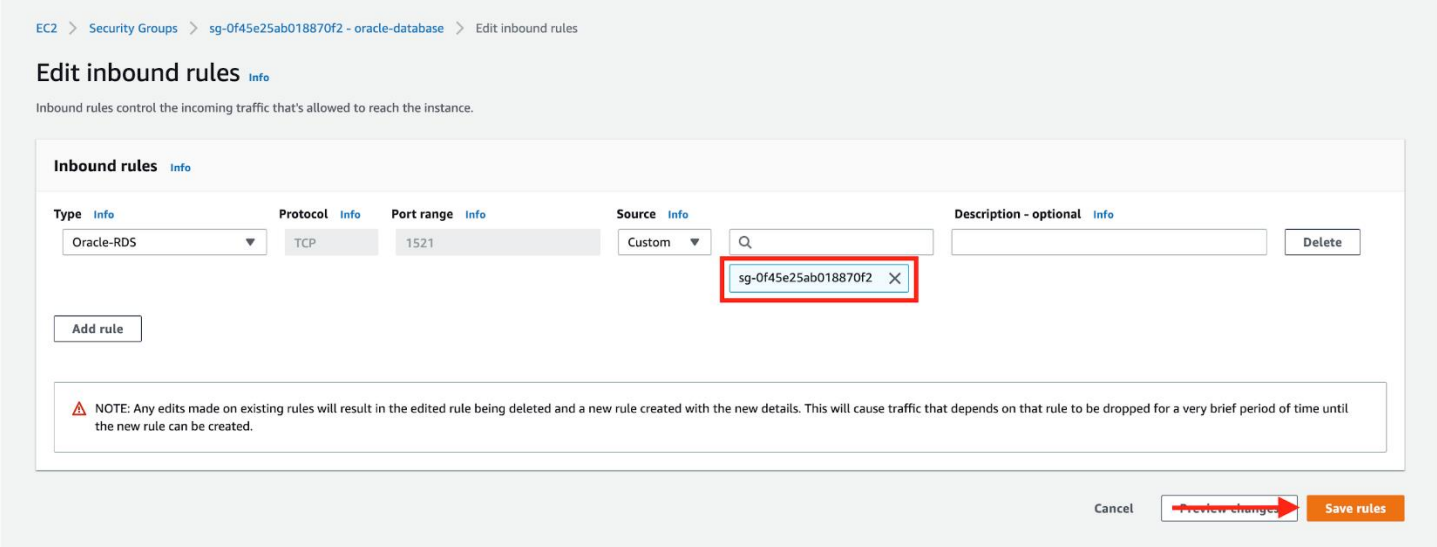
Cancel Create


DMS > Replication instances
Replication instances (1)
Actions
Create replication instance
Find replication instance
1
Name
Class
Status
Engine version
Availability zone
VPC
Public
Public IP address
Private IP address
myreplicationinstance
dms.t2.medium
Creating
3.3.1
vpc-0425287e
Yes

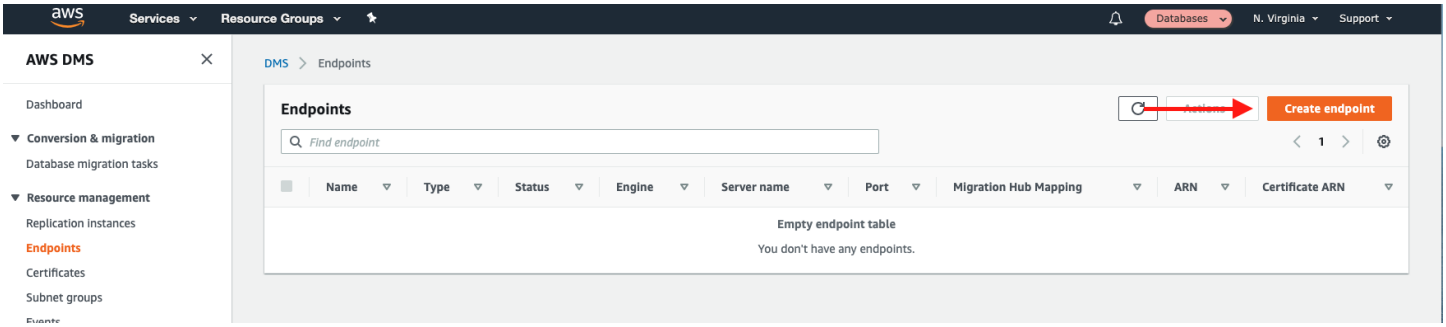

DMS > Replication instances
Replication instances (1)
Actions
Create replication instance
Find replication instance
1
Name
Class
Status
Engine version
Availability zone
VPC
Public
Public IP address
Private IP addr
my-replication-instance
dms.t2.medium
Available
3.3.1
us-east-1e
vpc-0425287e
No
-
172.31.55.218

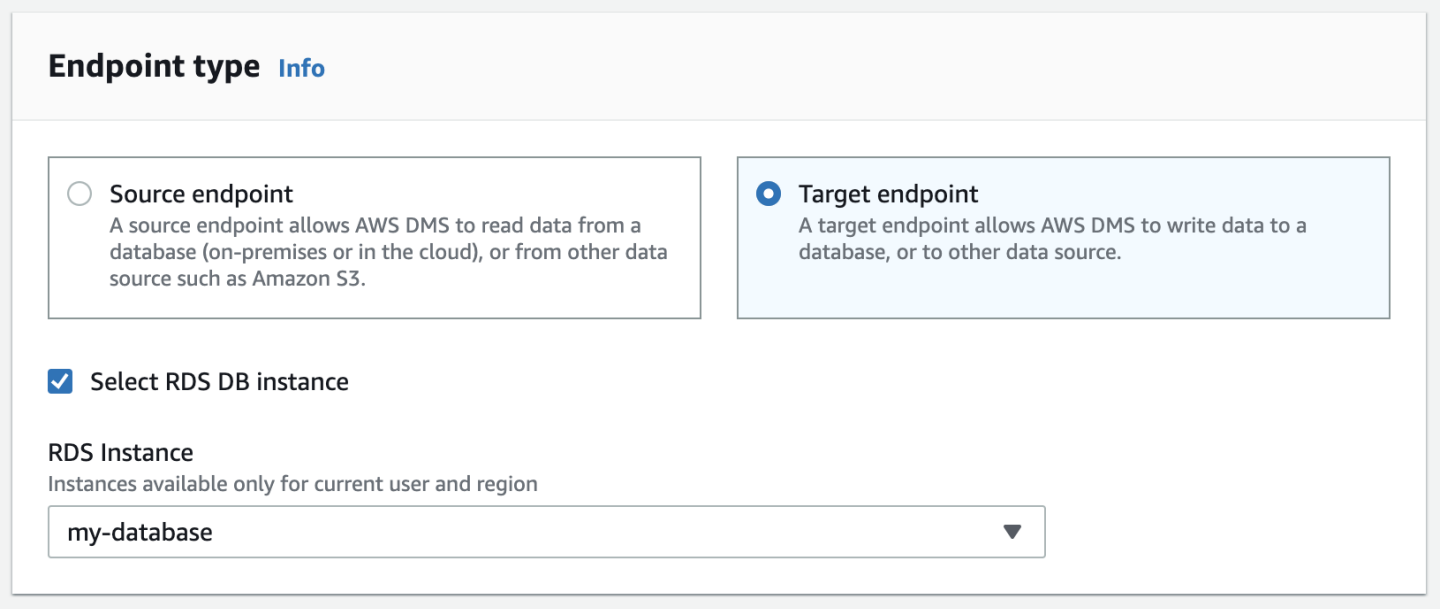

EC2 > Security Groups
Security Groups (3) Info
Actions
Create security group
Filter security groups
1
Security group ID
Security group name
VPC ID
Description
Owner
Inbound rules count
sg-0cef058753f829984
source-oracle-database
vpc-0425287e
Created by RDS manag...
905517915310
1 Permission entry
sg-0f45e25ab018870f2
oracle-database
vpc-0425287e
Created by RDS manag...
905517915310
1 Permission entry
sg-634d9e42
default
vpc-0425287e
default VPC security gr...
905517915310
1 Permission entry


EC2 > Security Groups > sg-0f45e25ab018870f2 - oracle-database
sg-0f45e25ab018870f2 - oracle-database
Delete security group
Copy to new security group
Details
Security group name
oracle-database
Security group ID
sg-0f45e25ab018870f2
Description
Created by RDS management console
VPC ID
vpc-0425287e
Owner
905517915310
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Tags
Inbound rules
Edit inbound rules
Type
Protocol
Port range
Source
Description - optional
Oracle-RDS
TCP
1521
97.119.243.198/32
-



➢ Creating Endpoints in AWS DMS





Endpoint configuration

Endpoint identifier **Info**
A label for the endpoint to help you identify it.
my-database

Target engine
The type of database engine this endpoint is connected to.
oracle

Server name
my-database.cv6ykrzkgphs.us-east-1.rds.amazonaws.com

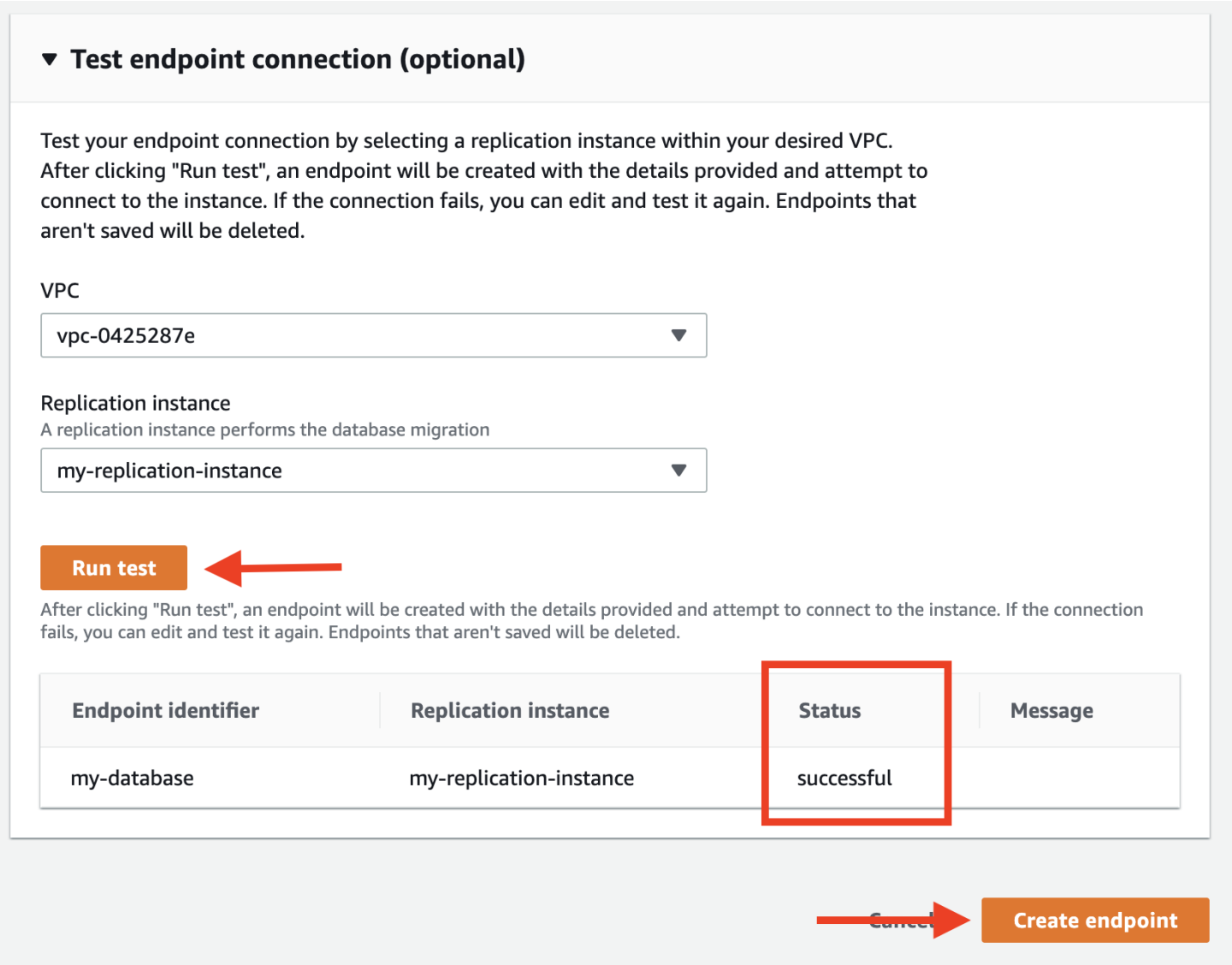
Port
The port the database runs on for this endpoint.
1521

Secure Socket Layer (SSL) mode
The type of Secure Socket Layer enforcement
none

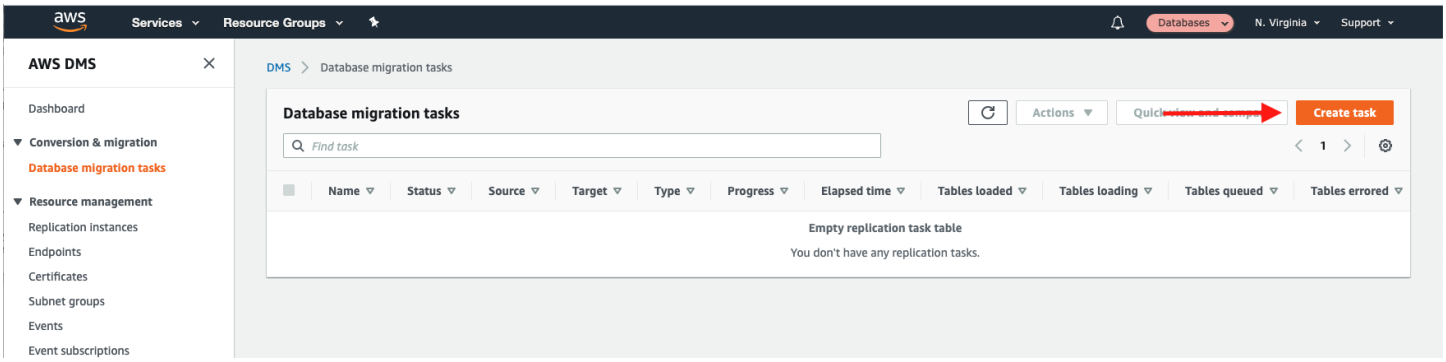
User name **Info**
admin

Password **Info**
••••••••

SID/Service name
Use Service name or SID as applicable. Use DB name in case of RDS for Oracle
ORCL



➢ Creating a Replication task in DMS



Create database migration task

Task configuration

Task identifier

oracle-migration

Replication instance

my-replication-instance - vpc-0425287e

Source database endpoint

source-oracle

Target database endpoint

my-database

Migration type Info

Migrate existing data

When switching database engines, the AWS Schema Conversion Tool can automatically convert your database schema and code to the engine of your choice. Click here to find out more. Learn more

Start task on create

Table mappings

Editing mode

- **Guided UI** — Set up your table mapping rules using a step-by-step guided interface.
- **JSON editor** Learn more — Enter your table mapping rules directly, in JSON format.

Specify at least one selection rule with an include action. After you do this, you can add one or more transformation rules.

Selection rules

Choose the schema and/or tables you want to include with, or exclude from, your migration task. Info

Add new selection rule

where **schema name** is like '%' and **table name** is like '%', include

Schema

Enter a schema

Schema name

Use the % character as a wildcard

ADMIN

Table name

Use the % character as a wildcard

%

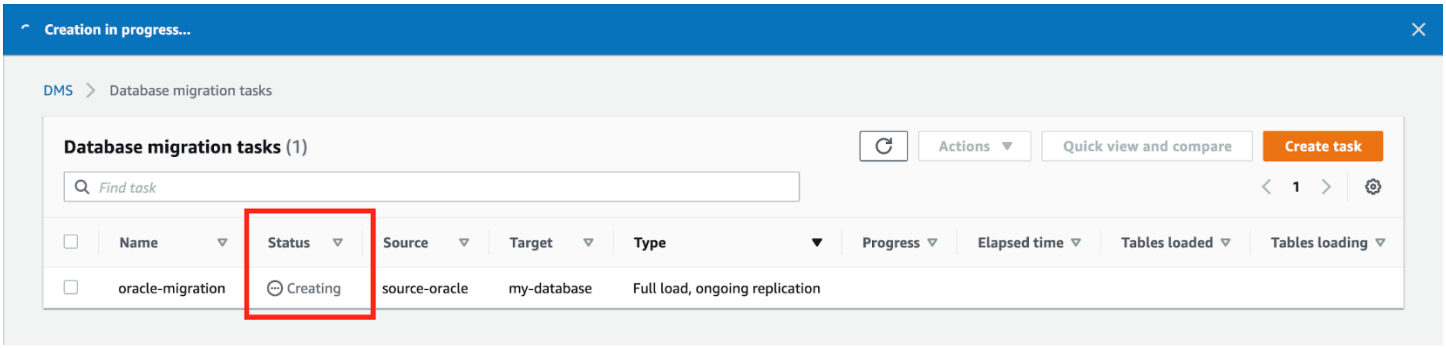
Action

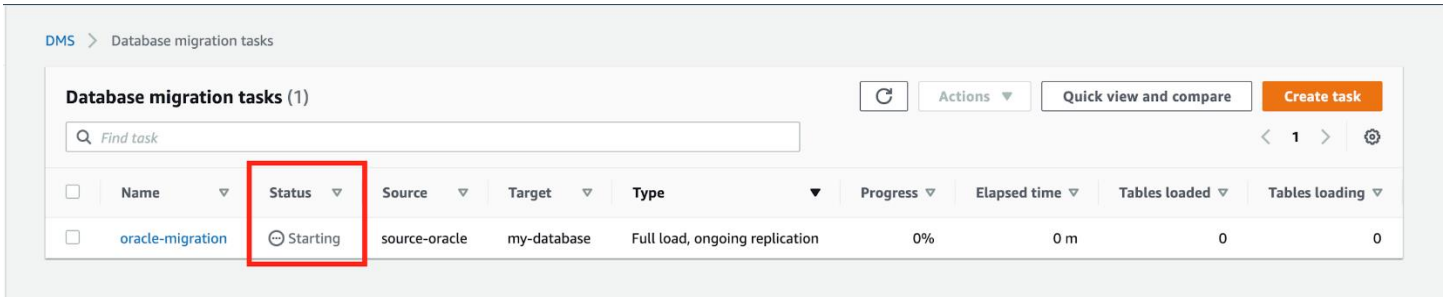
Choose "Include" to migrate your selected objects, or "Exclude" to ignore them during the migration.

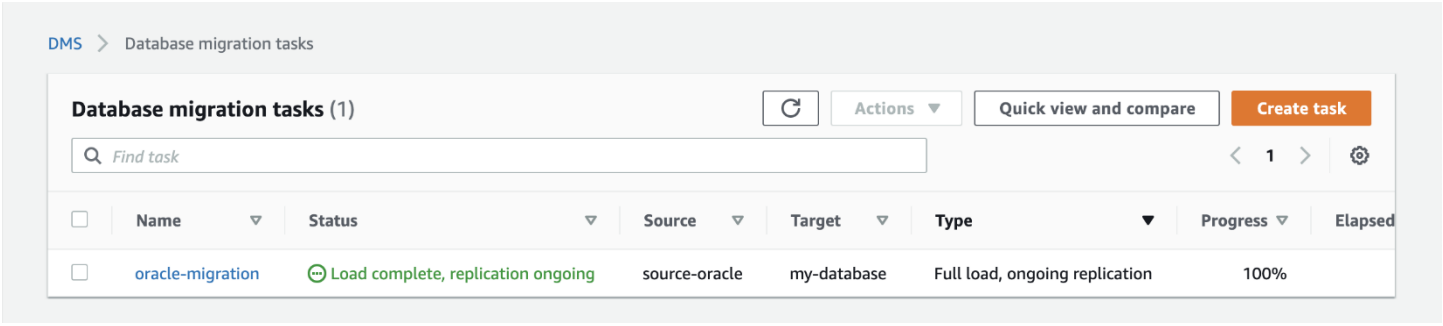
Include

Source filters Info

Add column filter




DMS > Database migration tasks
Database migration tasks (1)
Actions
Quick view and compare
Create task
Find task
1
Name
Status
Source
Target
Type
Progress
Elapsed time
Tables loaded
Tables loading
oracle-migration
Starting
source-oracle
my-database
Full load, ongoing replication
0%
0 m
0
0


DMS > Database migration tasks
Database migration tasks (1)
Actions
Quick view and compare
Create task
Find task
1
Name
Status
Source
Target
Type
Progress
Elapsed
oracle-migration
Load complete, replication ongoing
source-oracle
my-database
Full load, ongoing replication
100%